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PACKET DECODER LAB

Decoding Traffic: A Hands-On Guide to Transport-Layer Protocols

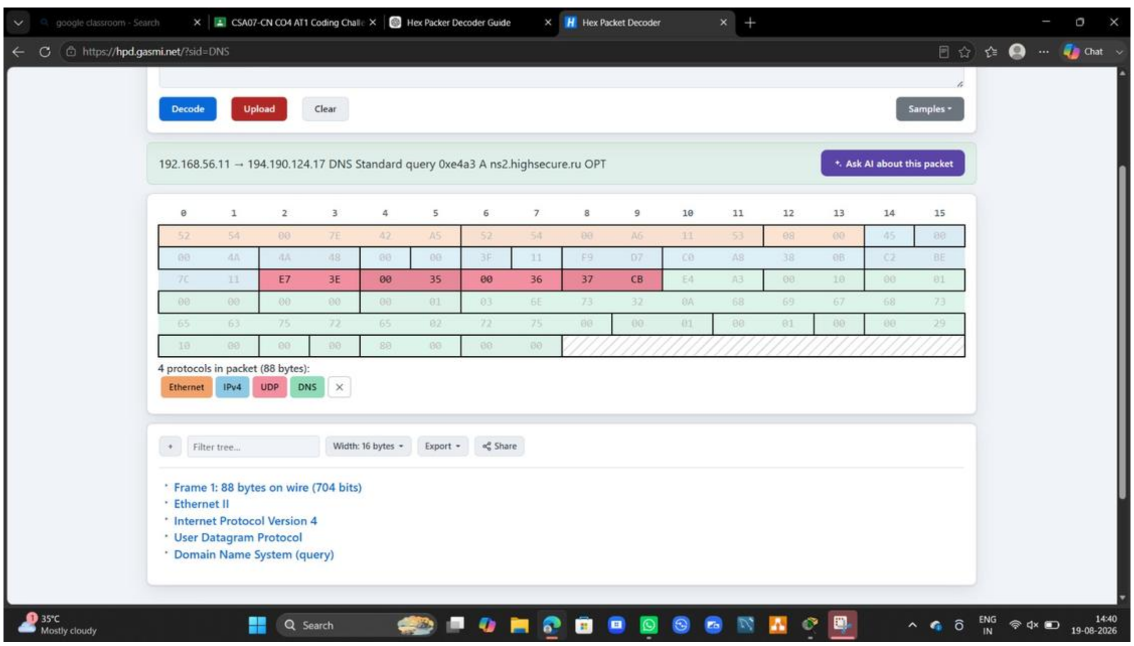
Ten practical exercises for breaking down UDP, TCP, and QUIC packets at the byte level.

SET 1 Dissecting UDP Packets

Three UDP packets examined field by field.

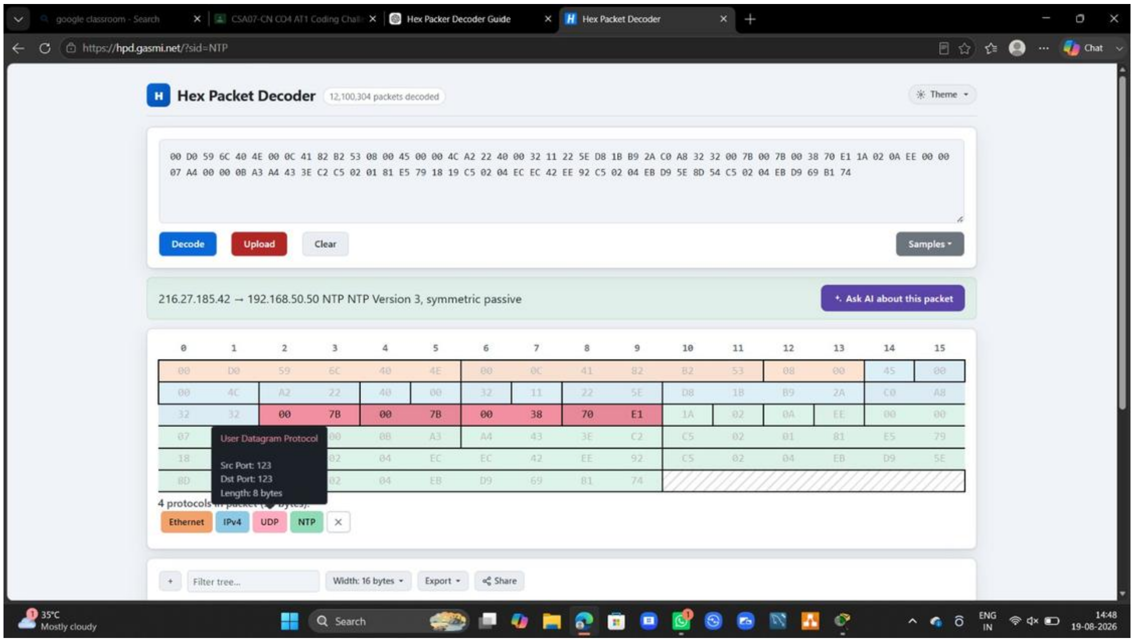
Exercise 1 — DNS Request

Host 192.168.56.11 issues a DNS lookup for ns2.highsecure.ru, directed at 194.190.124.17.



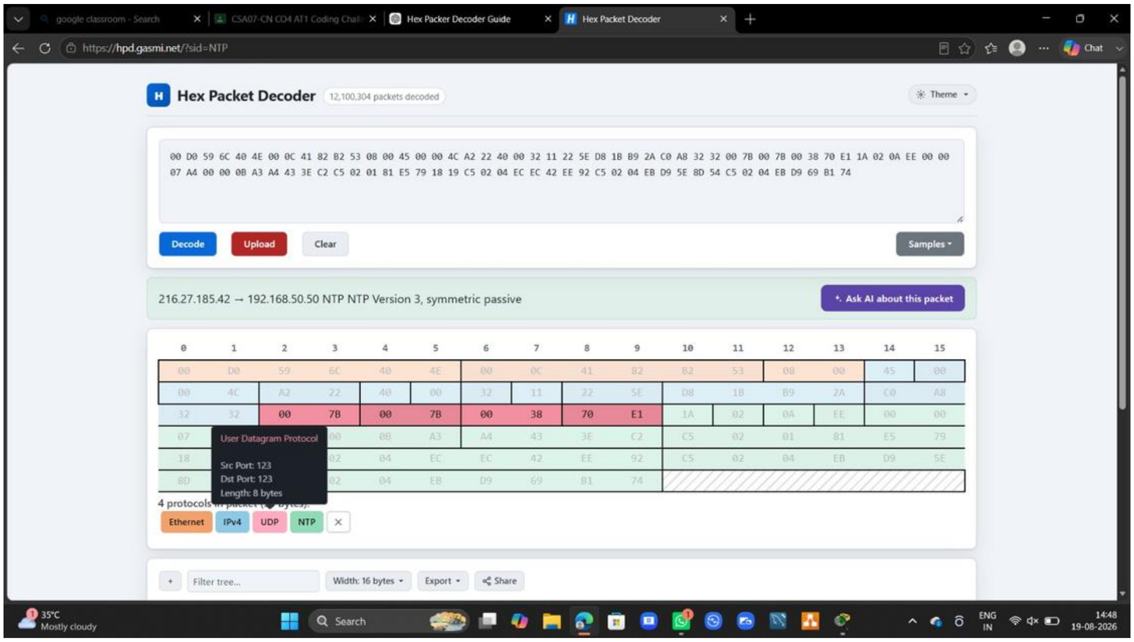
Exercise 2 — NTP Response

216.27.185.42 sends a time-sync reply back to 192.168.50.50. Notice both endpoints use port 123.



Exercise 3 — NTP Response, UDP Header Detail

The same packet as above, now with the UDP header opened up to reveal the length field.

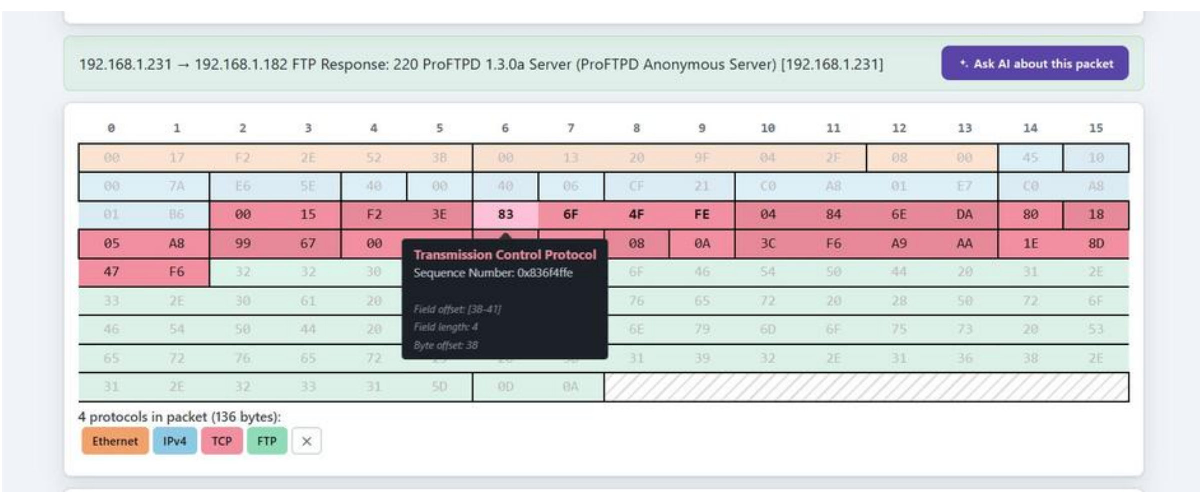


SET 2 Tracing a TCP Session

An FTP server response transported over TCP, peeled apart layer by layer.

Exercise 4 — Sequence Number

192.168.1.231 delivers the FTP welcome message to 192.168.1.182, with the sequence number called out.



Exercise 4 (cont.) — Acknowledgment Number

The identical packet, this time with the acknowledgment number highlighted.

Exercise 5 — IPv4 Header

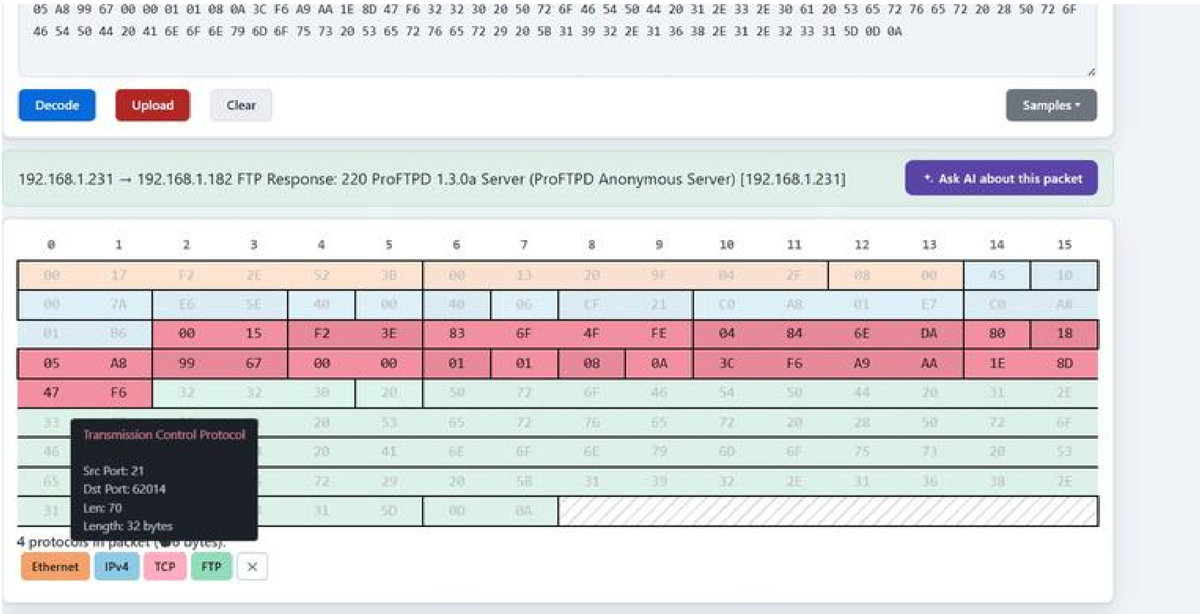
Moving up a layer on the same packet: the source and destination IP addresses.

SET 3 Comparing UDP and TCP

The same packets revisited, spotlighting different fields each round.

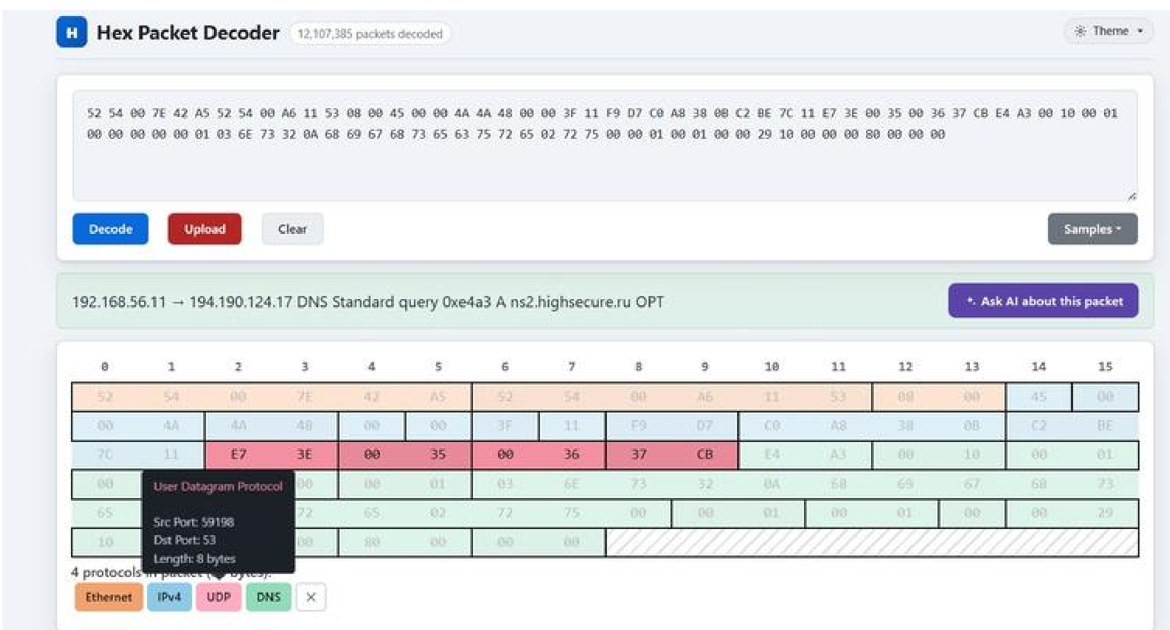
Exercise 6 — TCP Ports and Length

The FTP response again, this time showing source port, destination port, and segment length together.



Exercise 7a — UDP Ports

Back to the DNS request from Exercise 1 — the UDP header expanded to show the port pairing (59198 → 53).

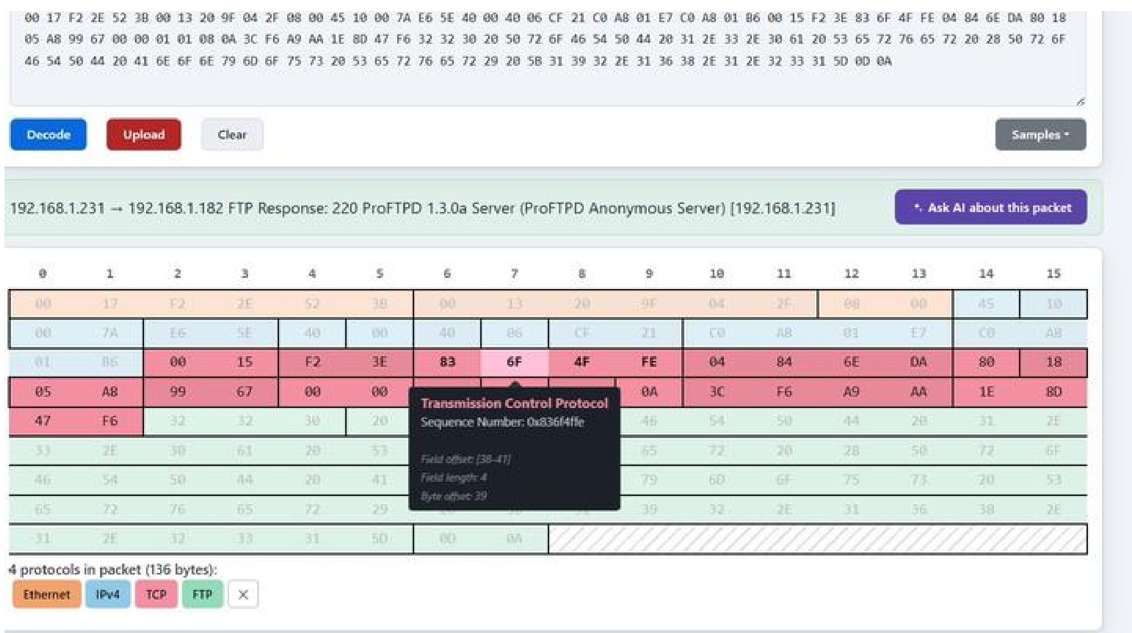


Exercise 7b — TCP Ports

The FTP response's TCP header shown for side-by-side comparison.

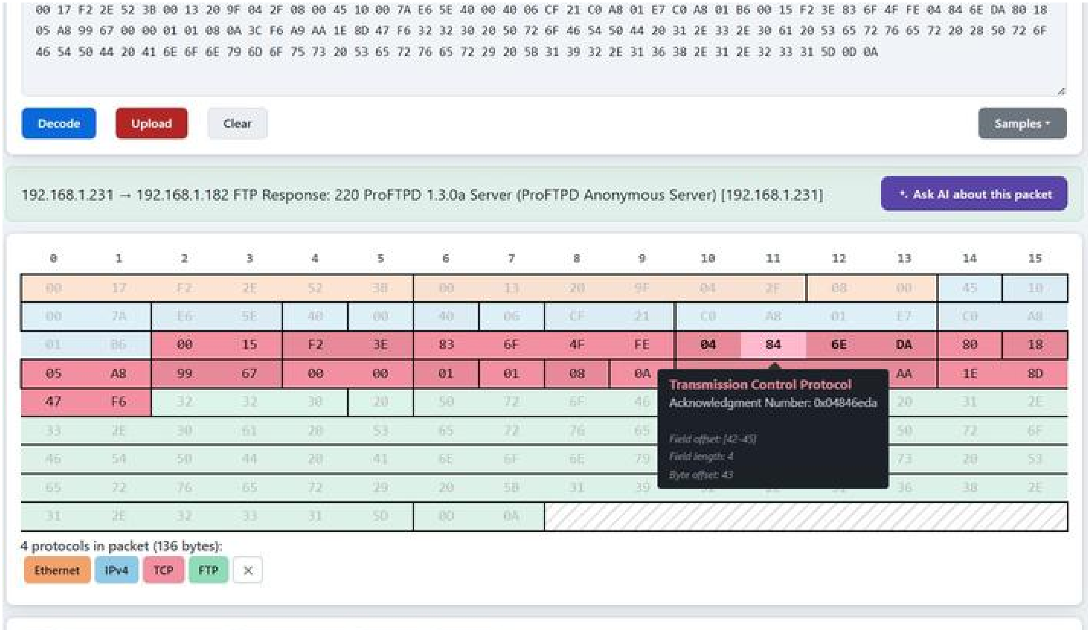
Exercise 8a — Sequence Number (revisited)

The FTP segment once more, this time pairing the sequence number with its byte offset.



Exercise 8b — Acknowledgment Number (revisited)

Same segment, now showing the acknowledgment number alongside its byte offset.

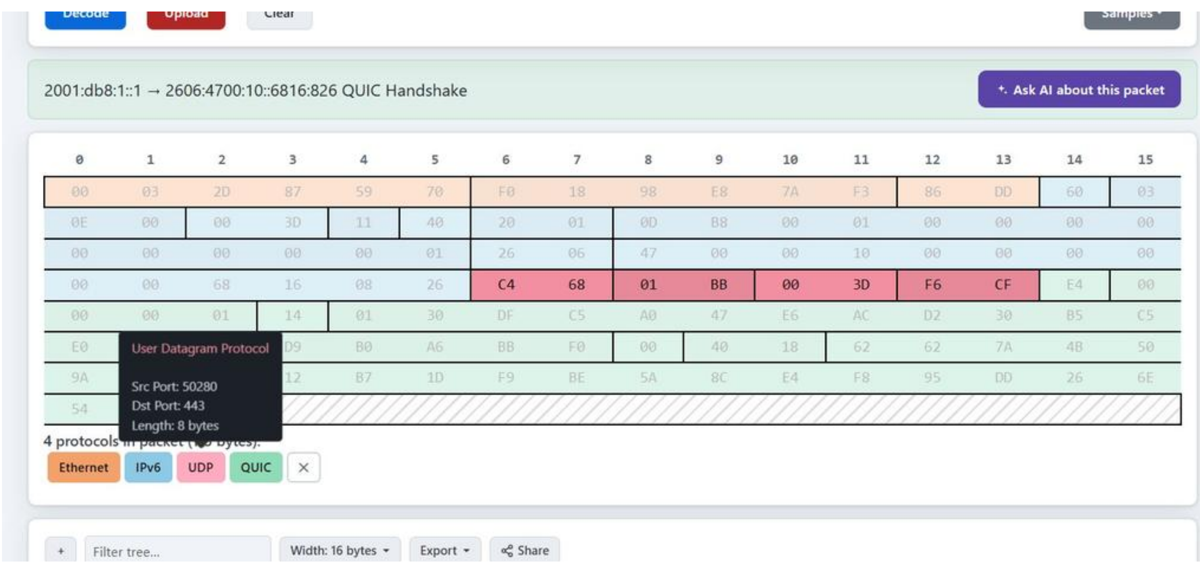


SET 4 Past UDP and TCP: Exploring QUIC

A QUIC connection handshake traveling over UDP on IPv6.

Exercise 10 — QUIC Handshake

Endpoints 2001:db8:1::1 and 2606:4700:10::6816:826 initiate a QUIC handshake, with the UDP header fields shown.



Exercise 10 (cont.) — Complete Frame

The entire 115-byte frame broken down across all layers: Ethernet, IPv6, UDP, and QUIC — checksum field highlighted.